

DNS UDP Scanning – Penetration Testing Assessment

Question: Analyze why a penetration tester chooses UDP scanning for a DNS server.

Answer

A penetration tester chooses UDP scanning for a DNS server because DNS commonly uses **UDP port 53**. UDP scanning helps determine whether the DNS service is exposed to the network and how it responds to UDP traffic.

Key Reasons

1. DNS Uses UDP Port 53

Most normal DNS queries and responses use UDP port 53. UDP scanning helps determine whether this service is open and reachable.

2. Identifies Exposed Services

A UDP scan can reveal whether a DNS service is exposed to the Internet or an internal network.

3. Detects Misconfigurations

Testing an authorized DNS server can help identify unsafe configurations, such as unnecessarily open recursive DNS service.

4. Provides Additional Information

DNS responses can help an assessor understand how the service behaves and identify useful service information.

5. Complements TCP Scanning

DNS can also use TCP, particularly for zone transfers and larger responses. Therefore, UDP scanning complements TCP scanning.

Conclusion

UDP scanning is important when assessing DNS because it checks the protocol normally used for DNS queries, identifies exposure, and can reveal configuration weaknesses that may not be discovered through TCP scanning alone. In an authorized assessment, UDP and TCP results should be correlated before drawing conclusions.

Kali Linux / Nmap Output

```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sU -p 53 8.8.8.8  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 11:25 IST  
Nmap scan report for dns.google (8.8.8.8)  
Host is up (0.029s latency).  
  
PORT      STATE      SERVICE  
53/udp    open|filtered domain  
  
Nmap done: 1 IP address (1 host up) scanned in 2.18 seconds  
veeravinod45@kali:~$
```

Illustrative Nmap output for an authorized DNS assessment. The result shown is **open|filtered**, which means filtering may prevent Nmap from determining the exact UDP port state.